

Radian measure



- 1
 - a State the multiplier to convert an angle from degrees to radians.
 - b State the multiplier to convert an angle from radians to degrees.

- 2 Convert the following to radians, correct to two decimal places where applicable:
 - a 360°
 - b 180°
 - c 90°
 - d 225°
 - e -300°
 - f 112°
 - g 29°
 - h 161.17°

- 3 Convert the following to degrees:
 - a $\frac{\pi}{3}$
 - b $\frac{\pi}{6}$
 - c $\frac{\pi}{4}$
 - d $\frac{3\pi}{2}$
 - e $\frac{5\pi}{4}$
 - f $-\frac{2\pi}{3}$
 - g $-\frac{5\pi}{3}$
 - h 4.2 radians

- 4 Determine whether the following statements are true or false. If it is false, correct the statement.
 - a If $180^\circ = \pi$ radians, then 90° must be equal to $\frac{\pi}{2}$ radians.
 - b 60° must be equal to $\frac{\pi}{6}$ radians, because it is $\frac{1}{6}$ of 360° .
 - c 30° must be equal to $\frac{\pi}{3}$ radians, because it is half of 60° .
 - d 210° must be equal to $\frac{7\pi}{6}$ radians, because it is seven groups of 30° .
 - e 45° must be equal to $\frac{\pi}{4}$ radians, because it is half of 90° .
 - f 225° must be equal to $\frac{7\pi}{4}$ radians, because it is seven groups of 45° .

Coterminal angles

- 5 Consider the angle $\frac{5\pi}{8}$.
 - a State the measure of the coterminal angle to $\frac{5\pi}{8}$ that is between 2π and 4π .
 - b State the measure of the coterminal angle to $\frac{5\pi}{8}$ that is between 0 and -2π .

6 Consider the angle $\frac{9\pi}{8}$.



- a State the measure of the coterminal angle to $\frac{9\pi}{8}$ that is between 2π and 4π .
- b State the measure of the coterminal angle to $\frac{9\pi}{8}$ that is between 0 and -2π .

7 Consider the angle $-\frac{2\pi}{3}$.

- a State the measure of the coterminal angle to $-\frac{2\pi}{3}$ that is between 0 and 2π .
- b State the measure of the coterminal angle to $-\frac{2\pi}{3}$ that is between -2π and -4π .

8 Consider the angle $\frac{4\pi}{3}$.

- a In which quadrant is the terminal side of this angle?
- b Determine the measure of the acute angle between the terminal side of $\frac{4\pi}{3}$ and the x -axis.
- c Now, what acute angle in the first quadrant can $\frac{4\pi}{3}$ be related to?

9 Consider the angle $\frac{8\pi}{7}$.

- a In which quadrant is the terminal side of this angle?
- b Determine the measure of the acute angle between the terminal side of $\frac{8\pi}{7}$ and the x -axis.
- c Now, what acute angle in the first quadrant can $\frac{8\pi}{7}$ be related to?

10 Consider an angle of 94° .

- a Find the measure of the smallest positive angle that is coterminal with 94° .
- b Find the first negative angle measure that is coterminal with 94° .
- c State which quadrant does 94° lie in.

11 Consider an angle of -58° .



- a Find the measure of the smallest positive angle that is coterminal with -58° .
- b Find the measure of the smallest negative angle that is coterminal with -58° .
- c State which quadrant -58° lies in.

12 Find the measure of the smallest positive angle that is coterminal with each of the following angles:

- a 489°
- b -331°

13 Find the measure of the smallest positive angle that is coterminal with each of the following angles, leaving your answers in radians:

- a $\frac{9\pi}{4}$
- b $-\frac{11\pi}{3}$

14 For each of the following, state the expression in terms of n , where n represents any integer, that generates all angles coterminal with the given angle:

- a $240^\circ + \mathbb{I}$
- b $-90^\circ + \mathbb{I}$

15 For each of the following, state the expression in terms of n , where n represents any integer, that generates all angles coterminal with the given angle:

- a $\frac{\pi}{2} + \mathbb{I}$
- b $-\frac{\pi}{2} + \mathbb{I}$

Reference angles

16 Find the measure of the reference angle for each of the following angles:

- a 111°
- b 197°
- c 275°

17 For each of the following angles:

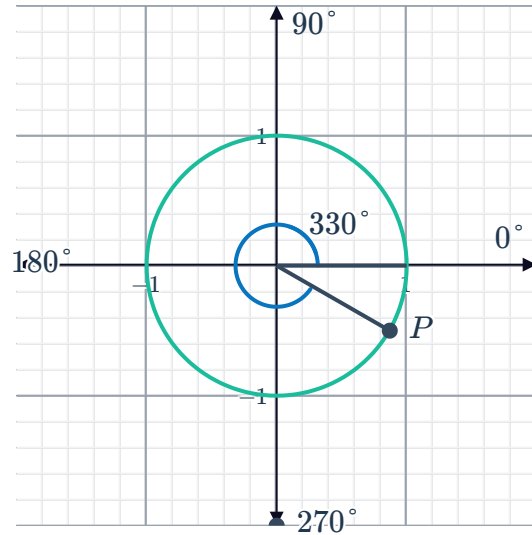
- i Find the measure of the coterminal angle that is between 0° and 360° .
- ii Find the measure of the reference angle.

- a -167°
- b 519°

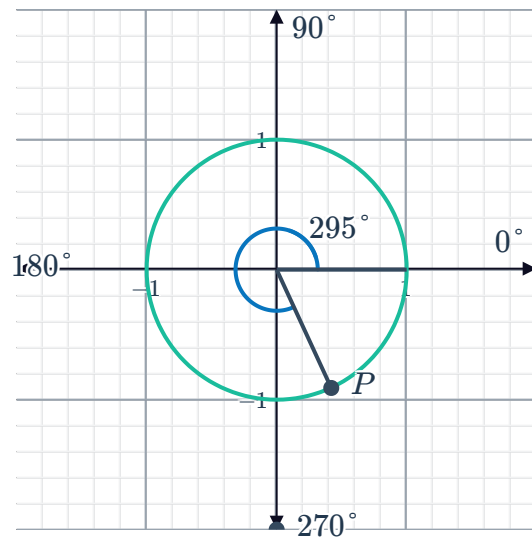
- 18** Point P on the unit circle shows a rotation of 330° .



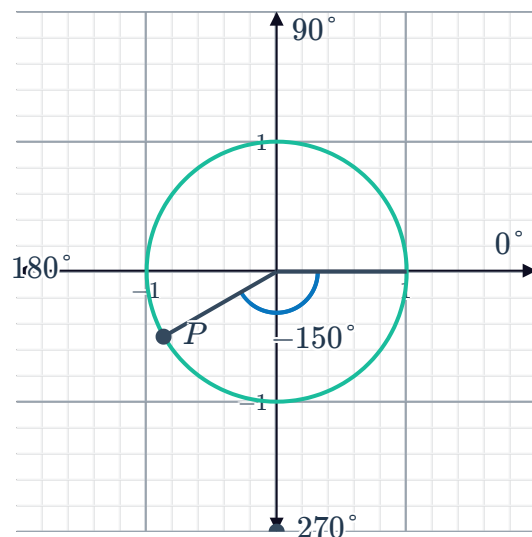
Find the measure of the related acute angle in the first quadrant.



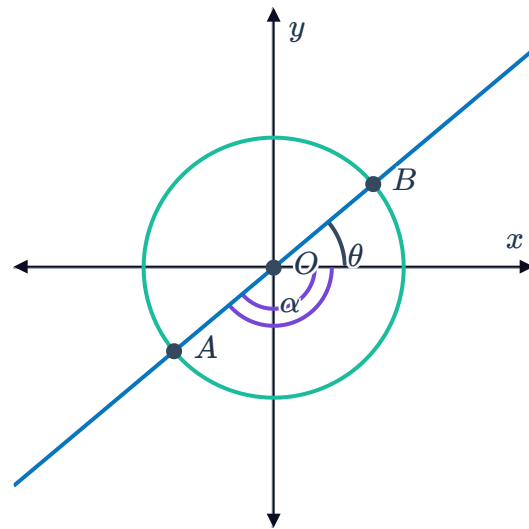
- 19** Point P on the unit circle shows a rotation of 295° .
Find the measure of the related acute angle in the first quadrant.



- 20** Point P on the unit circle shows a rotation of -150° .
Find the measure of the related acute angle in the first quadrant.

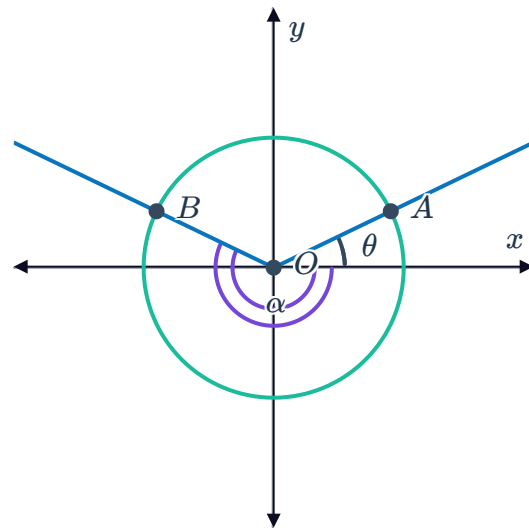


- 21** The graph shows a line through two points on the unit circle, A and B , and the origin, O . The line segment OB forms an angle of $\theta = \frac{\pi}{13}$ with the positive x -axis. The angle between the line segment OA and the positive x -axis is α . Find the value of α when:



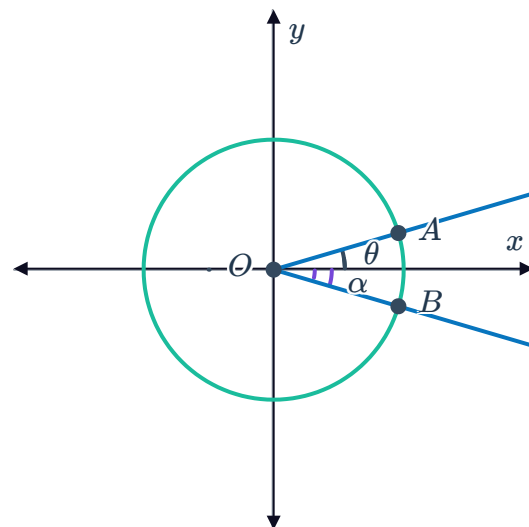
- a** $0 \leq \alpha < 2\pi$ **b** $-2\pi < \alpha \leq 0$

- 22** Points A and B are two points on the unit circle centered at the origin, O . The line segment OA forms an angle of $\theta = \frac{\pi}{7}$ with the positive x -axis, and the line passing through A and B is parallel to the x -axis. The angle between the line segment OB and the positive x -axis is α . Find the value of α when:



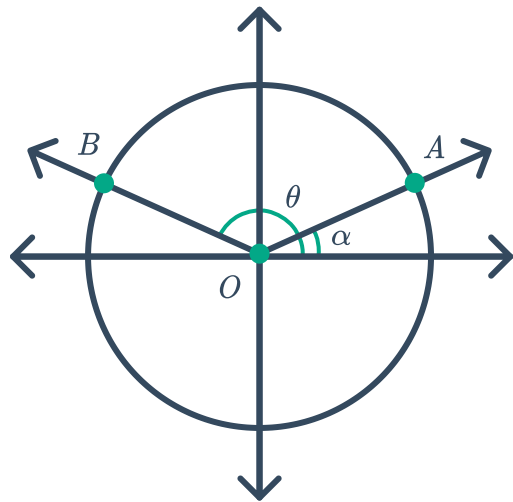
- a** $0 \leq \alpha < 2\pi$ **b** $-2\pi < \alpha \leq 0$

- 23** Points A and B are two points on the unit circle centered at the origin, O . The line segment OA forms an angle of $\theta = \frac{\pi}{11}$ with the positive x -axis, and the line passing through A and B is parallel to the y -axis. The angle between the line segment OB and the positive x -axis is α . Find the value of α when:

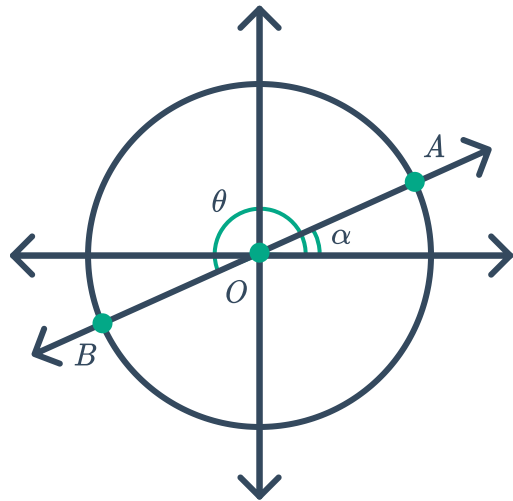


- a** $0 \leq \alpha < 2\pi$ **b** $-2\pi < \alpha \leq 0$

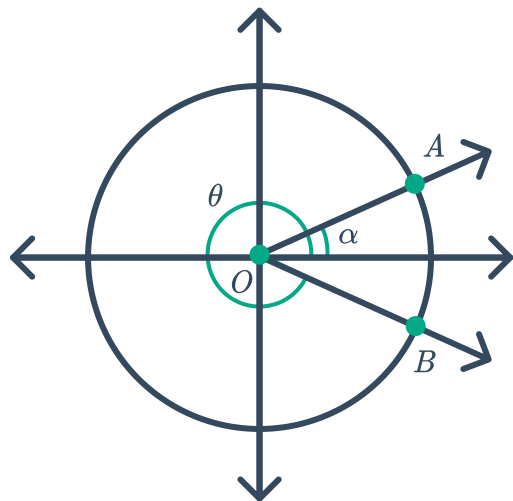
- 24** Consider the unit circle shown where points A and B have the same y -coordinates. Suppose that $\theta = \frac{10\pi}{11}$. State the measure of the reference angle, α .



- 25** Consider the unit circle shown where the line through A and B passes through the origin, O . Suppose that $\theta = \frac{8\pi}{7}$. State the size of the reference angle, α .



- 26** Consider the unit circle shown where points A and B have the same x -coordinate. Suppose that $\theta = \frac{9\pi}{5}$. State the measure of the reference angle, α .



27 Find the measure of the reference angle for 

28

- a Find the measure of the coterminal angle to $-\frac{8\pi}{9}$ that is between 0 and 2π .
- b Now, find the measure of the reference angle for $-\frac{8\pi}{9}$.

29 Consider the acute reference angle with measure $\frac{\pi}{17}$.

State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:

a $\sin \theta = \sin \frac{\pi}{17}$

b $\sin \theta = -\sin \frac{\pi}{17}$

c $\cos \theta = \cos \frac{\pi}{17}$

d $\cos \theta = -\cos \frac{\pi}{17}$

e $\tan \theta = \tan \frac{\pi}{17}$

f $\tan \theta = -\tan \frac{\pi}{17}$

30 Consider the angle with measure $\frac{9\pi}{13}$.

a State the measure of the acute reference angle for $\frac{9\pi}{13}$.

b State all values of α over the domain $0 \leq \alpha \leq 2\pi$ for which:

i $\sin \alpha = \sin \frac{9\pi}{13}$

ii $\cos \alpha = \cos \frac{9\pi}{13}$

iii $\tan \alpha = \tan \frac{9\pi}{13}$

31 Consider the angle $\frac{18\pi}{17}$.

a State the measure of the reference angle for $\frac{18\pi}{17}$.

b State all values of α over the domain $0 \leq \alpha \leq 2\pi$ for which:

i $\sin \alpha = \sin \frac{18\pi}{17}$

ii $\cos \alpha = \cos \frac{18\pi}{17}$

iii $\tan \alpha = \tan \frac{18\pi}{17}$

32 Consider the angle $\frac{32\pi}{17}$.



- a State the measure of the reference angle for $\frac{32\pi}{17}$.
- b State all values of α over the domain $0 \leq \alpha \leq 2\pi$ for which:
- i $\sin \alpha = \sin \frac{32\pi}{17}$
- ii $\cos \alpha = \cos \frac{32\pi}{17}$
- iii $\tan \alpha = \tan \frac{32\pi}{17}$

33 Consider the angle $\frac{9\pi}{17}$.

- a State the measure of the reference angle for $\frac{9\pi}{17}$.
- b State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:
- i $\sin \theta = -\sin \frac{9\pi}{17}$
- ii $\cos \theta = -\cos \frac{9\pi}{17}$
- iii $\tan \theta = -\tan \frac{9\pi}{17}$

34 Consider the angle $\frac{19\pi}{17}$.

- a State the measure of the reference angle for $\frac{19\pi}{17}$.
- b State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:
- i $\sin \theta = -\sin \frac{19\pi}{17}$
- ii $\cos \theta = -\cos \frac{19\pi}{17}$
- iii $\tan \theta = -\tan \frac{19\pi}{17}$

35 Consider the angle $\frac{21\pi}{13}$.

- a State the measure of the reference angle for $\frac{21\pi}{13}$.
- b State all values of θ over the domain $0 \leq \theta \leq 2\pi$ for which:
- i $\sin \theta = -\sin \frac{21\pi}{13}$
- ii $\cos \theta = -\cos \frac{21\pi}{13}$
- iii $\tan \theta = -\tan \frac{21\pi}{13}$